

Richeek Das

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Research Interests

Event-based Vision, Self Supervised Representation Learning, Robotic Perception, Multimodal Sensing

Education

University of Pennsylvania

PhD in Computer and Information Science

Advisor: Prof. Pratik Chaudhari

Philadelphia, USA

2023 - Ongoing

Indian Institute of Technology Bombay

Bachelor of Technology with Honors in Computer Science and Engineering

Advisor: Prof. Preethi Jyothi

Thesis: Code-switched Text Modelling for Natural Language Understanding

Received the **Research Excellence Award** for undergraduate thesis

Mumbai, India

2019 - 2023

GPA: 9.62/10.0

Journal Papers

2. **Richeek Das**, Kostas Daniilidis, Pratik Chaudhari, **Fast Feature Field (F^3): A Predictive Representation of Events**, *in submission 2025*
1. **Richeek Das**, Aaron Jerry Ninan, Adithya Bhaskar, Ajit Rajwade, **Performance Bounds for LASSO under Multiplicative LogNormal Noise: Applications to Pooled RT-PCR testing**, *accepted at the Signal Processing Journal 2024*

Conference Papers

5. **Richeek Das**, Samuel Dooley, **Fairer and More Accurate Tabular Models Through NAS**, *accepted at the Algorithmic Fairness through the Lens of Time workshop of NeurIPS 2023*
4. **Richeek Das**, Sahasra Ranjan, Shreya Pathak, Preethi Jyothi, **Pretraining Techniques for Improved Code-switched Natural Language Understanding**, *accepted at the ACL 2023 (Association for Computational Linguistics)* - 🏆 **ACL Outstanding Paper Award (top 1%)**
3. Alex Markham, **Richeek Das**, Moritz Grosse-Wentrup, **A Distance Covariance-based Kernel for Nonlinear Causal Clustering in Heterogeneous Populations**, *accepted at the CLear 2022 (1st conference on Causal Learning and Reasoning)*
2. Alexander Erlei, **Richeek Das**, Lukas Meub, Avishek Anand, Ujwal Gadiraju, **For What It's Worth: Humans Overwrite Their Economic Self-interest to Avoid Bargaining With AI Systems**, *accepted at the ACM CHI 2022 (Conference on Human Factors in Computing Systems)*
1. Ashish Tiwari, **Richeek Das**, Shanmuganathan Raman, **Exploring Deeper Graph Convolutions For Semi-Supervised Node Classification**, *accepted at the IEEE ICASSP 2022 (International Conference on Acoustics, Speech, and Signal Processing)*

Research and Work Experience

Representation Learning for Event-based Vision | UPenn, GRASP Lab

Guide: Prof. Pratik Chaudhari, Prof. Kostas Daniilidis

Jul 2024 - Sep 2025

- Developed a mathematical argument and efficient algorithms for building representations event camera data—retains information about the structure and motion of the scene, robust to noise and event rates (120 Hz at HD resolution)
- SOTA performance on optical flow, segmentation and monocular depth on data from 3 platforms (car, quadruped robot, drone), different lightning (daytime, nighttime) and environments (indoors, outdoors, urban, off-road)

Extracting Scene Material Properties from Radar Signals | UPenn, WAVES Lab 🦾

Guide: Prof. Mingmin Zhao

Aug 2023 - Jan 2024

- Built hardware – a handheld device with synchronized iPad LiDAR, TI mmWave radar, camera and an onboard computer to capture data in real 3D indoor scenes with diverse materials and geometries
- Built software – a differentiable radar simulator to model wave propagation, antenna patterns and ray tracing in 3D scenes, and a novel SSL framework for extracting material properties using the physics of wave-material interactions

Real-time Multirotor Simulation Software | UPenn, GRASP Lab

Guide: Prof. Pratik Chaudhari

Mar 2025 - Ongoing

- A fast and modular simulation environment for multirotors with onboard IMUs, RGB, event, depth and semantic cameras. Playground to test real high-speed agile flight algorithms in simulation – control, planning, perception
- Built CUDA modules for event generation – efficiently integrating Habitat with RotorPy (~2400 FPS on RTX4090)

Fairer and More Accurate Tabular Models Through NAS | Abacus.AI 🦾

Guide: Dr. Samuel Dooley

May 2023 - Jul 2023

- Proposed a novel multi-objective NAS technique to utilize the implicit fairness in model architectures
- Performed extensive experiments over model search spaces and proved the existence of architectures that are inherently fair and accurate – enabling us to search for these models which surpass current SOTA debiasers

Neural Architecture Search Toolkit for Computer Vision | Sony AI, Japan 🦾

Guide: Takuya Narihira, Hsingying Ho

May 2022 - Jul 2022

- Proposed a novel Once-For-All based NAS framework for Semantic Segmentation in NNabla NAS, with plug-and-play features for building dynamic sub-networks with differing hardware-constraints without any re-training
- Adapted the method to DeepLabv3+ with dynamic sub-networks – 15% increase in mIoU and 2× latency reduction

Code-Switched Natural Language Understanding | Google India + IIT Bombay, CSALT 🦾

Guide: Prof. Preethi Jyothi

Jul 2022 - Jan 2023

- Implemented intelligent masking strategies for MLM pretraining and built architectures specific to code-switched tasks with significant improvements in downstream Question Answering and Sentiment Analysis
- Built a generalized framework to adapt existing language translation models for low-resource code-switched text generation with multiple constraints: formality, politeness, toxicity, semantic similarity and more

Kernel methods for Non-linear Causal Clustering | Universität Wien, Neuroinformatics Lab 🦾

Guide: Prof. Moritz Grosse-Wenstrup

May 2021 - Oct 2021

- Implemented a distance covariance-based kernel to measure the similarity between underlying nonlinear causal structures of samples – clustering according to their parent causal structures in heterogeneous populations
- Simulated causal datasets with non-linear relations to numerically evaluate the performance bounds of the distance covariance-based dependence contribution kernel and compare it with standard RBF and Polynomial kernels

Belief Elicitation on the Impact of AI Systems | TU Delft, Delft AI Labs 🦾

Guide: Prof. Ujwal Gadiraju

May 2021 - Oct 2021

- Implemented Binarized Scoring Rule based criterion for Belief Elicitation of user behaviour, presumptions and trust on the usage of Decision Support Systems (AI-System) for Algorithmic Bargaining
- Built and deployed a DRF backend, Angular frontend, PostgreSQL DB application coupled with Redis + Celery task management, on a Heroku + GitHub deployment pipeline to host 2700+ crowdsourcing submissions

Feature Gating for Deeper Graph Convolution Networks | IIT Gandhinagar, CVIG Lab

Guide: Prof. Shanmuganathan Raman

Dec 2020 - Jun 2021

- Introduced feature gating and formulated a heuristic to award importance scores to graph nodes and node features
- Proposed the use of identity mapping, a modified form of residual connection and feature gating to create deep GCN models which tackle oversmoothing and achieve SOTA results for semi-supervised node classification

Weighted LASSO under Multiplicative LogNormal Noise | IIT Bombay 🦾

Guide: Prof. Ajit Rajwade

Jan 2022 - Feb 2023

- Theoretically justified and interpreted the use of Tapestry Pooling across hospitals for pooled RT-PCR group testing
- Proposed a novel Weighted LASSO algorithm with data-dependent weights to perform sparse signal recovery under multiplicative LogNormal noise (example Gaussian) – performance backed using simulations on real RT-PCR data

Selected Projects

- Video from a Single Exposure Coded Snapshot**  Spring 2021
- Implemented a MATLAB solution for coded aperture compressive temporal imaging to recover a sequence of frames from a single coded-snapshot to achieve temporal gains in video acquisition without spatial compromise
- Compressed Sensing in Tomographic Reconstruction**  Spring 2021
- Implemented compressed sensing solution for tomographic reconstruction of brain MRI scans with small number of measurement angles – performed coupled reconstruction assuming similarity of consecutive acquisitions
- Cluster Monitoring and Alert System** Spring 2022
- Built a web-app with DRF, Angular and InfluxDB – telegraf servers on host machines to perform cluster profiling and alert low-resource warnings in real-time using socket-servers, based on thresholds set by users
- Branch Predictors for trace-based Simulators** Autumn 2021
- Implemented branch predictors TAGE and L-TAGE in ChampSim, and performed extensive comparisons with Bi-modal and Hashed Perceptron on the metrics of tag width, MPKI, TAGE Table Size and history length
- Compiler for C-like Language** Spring 2022
- Constructed a compiler handling a subset of the C language, designing a recursive descent parser using lex and yacc – supports type inference, semantic checks and translation of AST to linear three-address codes
- Extending xv6 Operating System** Autumn 2021
- Extended the xv6 OS with syscalls for demand paged memory allocation and custom fork implementations. Implemented thread synchronization, semaphores and a simple linux-based disk emulated filesystem in C

Selected Achievements and Awards

- Received the **Paul S. Darnell** named CIS PhD fellowship at UPenn. (2023)
- Received the **Thomas Doobie, Class of 1974 Research Excellence Award** for B.Tech Thesis. (2023)
- Among the select few undergrads to attend the **Google Research Week** hosted in Bangalore, India (2023)
- Secured a rank of **497** in JEE Main and **544** in JEE Advanced among **1.2 million** candidates (2019)
- Received the **INSPIRE** scholarship, awarded to **top 1%** of the **80k+** students in the **ISC** Exam (2019)
- Received the **KVPY Fellowship** for securing **77th** rank out of **150k+** candidates in all of India (2018)
- Secured **4th rank in India** in **ICSE** out of **180k+** candidates (2017)

Teaching Experience

- Introduction to Machine Learning** | *Prof. Biplob Banerjee, Prof. Manjesh Hanawal* Spring 2023
- Teaching Practicum for TAs** | *Dept. of Computer Science and Engineering* Autumn 2022, Spring 2023
- Software Systems Lab (Excellence in CSE Taship Award)** | *Prof. Amitabha Sanyal* Autumn 2021
- Computer Networks** | *Prof. Vinay Ribeiro* Autumn 2022
- Computer Programming and Utilization** | *Prof. Kameswari Chebrolu* Spring 2021
- Engineering Drawing** | *Prof. Atul Sharma* Summer 2021